# BRAC UNIVERSITY

**Department of Computer Science & Engineering**

**Assignment 1**

**CSE 350: Digital Electronics and Pulse Techniques**

**Total Marks:** 30 `TSE, Spring 26.`

- Total no of questions are **four (4)**. Answer **them all**.
- Figure in bracket [ ] next to each question indicates marks for that question.

## ■ Question 1 *[CO3] [20 marks]*

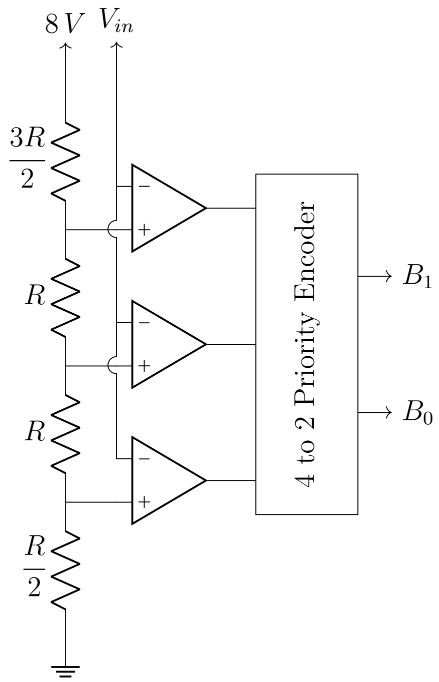


Figure 1: 2-bit Flash ADC

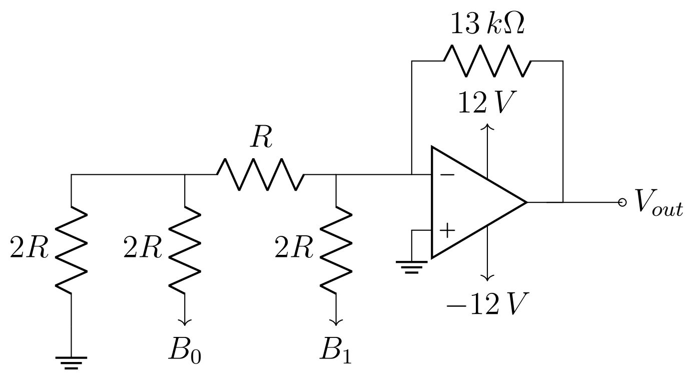


Figure 2: 2-bit R-2R Ladder DAC

$$V_{out} = -\frac{R_f}{2R}\left(V_{B_1} + \frac{V_{B_0}}{2}\right)$$

For both the circuits in the above figures, voltage of $B_1$ & $B_0$ is $5\,V$ when logic is **HIGH**, and $0\,V$, when logic is **LOW**. The output of the ADC is passed to the DAC as input.

(a) Make a table listing the **quantization intervals** of the ADC shown in figure 1, along with the respective **digital output**. [4]

(b) If the maximum digital input to the DAC, produces a $V_{out}$ of $\boldsymbol{-6.5\,V}$, find the value of the resistor 'R' in figure 2. Then calculate the **step size** of the DAC, and list all the output voltages for the **remaining inputs**. [4]

(c) Using your results from (a) & (b), find the digital output of the ADC for $\boldsymbol{V_{in} = 5.5\,V}$ and use it to find the $\boldsymbol{V_{out}}$ of the DAC. [3]

Now consider a 2-bit **Dual Slope ADC** with a similar reference voltage as in figure 1, was used instead of the Flash ADC. It has a clock frequency of $\boldsymbol{2\,MHz}$.

(d) Find the input voltage $\boldsymbol{V_{in}}$ for which the conversion time was found to be $\boldsymbol{2.7\,\mu s}$. Also find the **maximum conversion time** of the ADC. [4]

(e) If for the Dual Slope ADC $\boldsymbol{V_{in} = 5.5\,V}$, find its digital output '$B_1B_0$'. Then find $\boldsymbol{V_{out}}$ if this was passed to the DAC in figure 2. Find the **%error** between the values of $V_{out}$ for '1(c)' & '1(e)'. [5]

## ■ Question 2 *[CO3] [15 marks]*

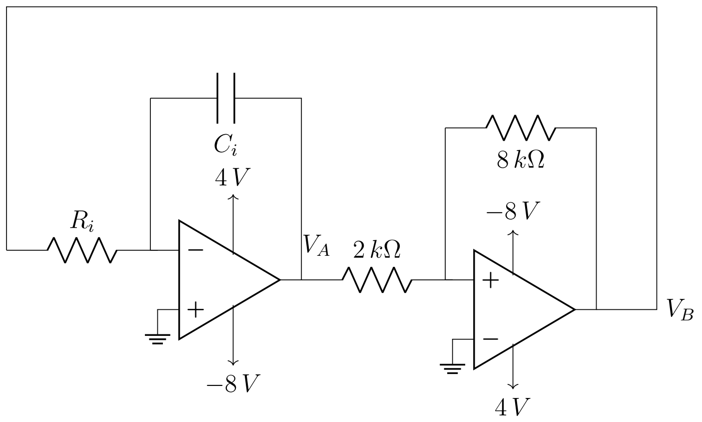


Figure 3: A Triangular Wave Generator

(a) For the triangular wave generated by the circuit in figure 3, prove that $\boldsymbol{f = \frac{8}{9R_iC_i}}$. [6]

(b) You have to design a **square wave generator** that can generate a square wave having duty cycle similar to the square wave at $V_B$. Find $V_H$, $V_L$ and draw the circuit given that, $R_1 = 10\,k\Omega$, $R_2 = 20\,k\Omega$, $R_x = 10\,k\Omega$, $C_x = 50\,nF$ and $f = 1\,kHz$. Here, $\tau = R_xC_x$ and $R_2$ is the feedback resistance. [*Hint: You may need to assume one of the bias voltages*] [9]

## ■ Question 3 *[CO3] [15 marks]*

The input to & output from a certain Schmitt Trigger is shown in figure 4.

(a) From figure 4, find the **bias voltages** & the **threshold voltages** of the Schmitt Trigger. Then draw its **VTC** (Voltage Transfer Characteristics) on the plot given in figure 5. [5]

(b) Find the **shift voltage**. Design the Schmitt Trigger by specifying the values of each component and drawing the full circuit. [5]

(c) If the connections of **input voltage** & **reference voltage** in the circuit of 'b' are swapped, find the new **thresholds**, **shift voltage** & the **hysteresis width**. What type of Schmitt Trigger is this modified circuit? [5]

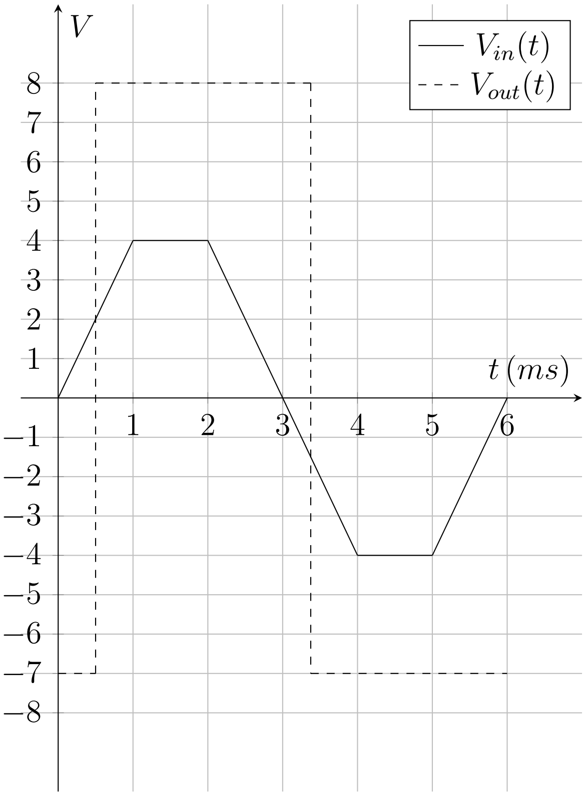


Figure 4: $V_{in}(t)\,\&\,V_{out}(t)$

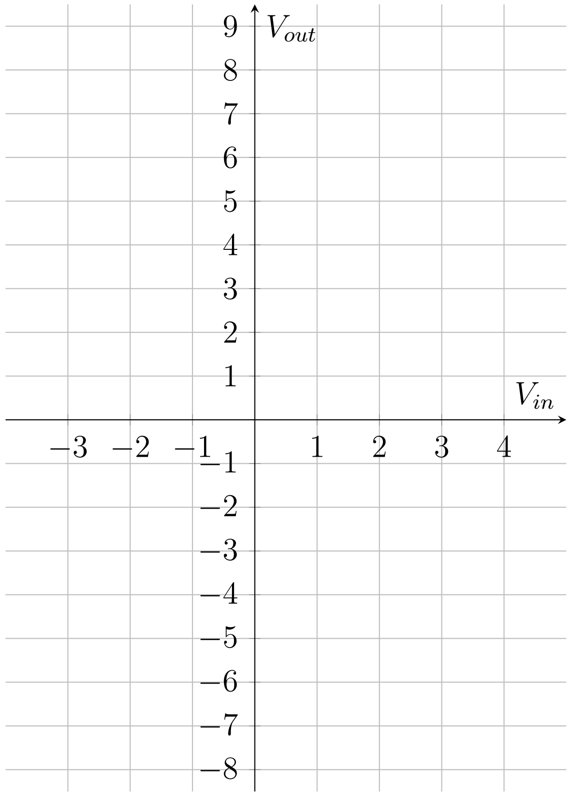


Figure 5: VTC of the Schmitt Trigger